

SAME SIZE ARTWORK
LEAFLET SIZE : 88 mm x 186 mm

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24 mm

SPACE FOR
PHARMACODE

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory.

MilicalTM
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Composition

Each film coated tablet contains:
Calcium Citrate USP 1000 mg
Vitamin D3 Granules
equivalent to Cholecalciferol 200IU

Description

A printed carton containing an insert & 3 printed blister strips, each containing 10, white capsule shaped film coated tablets with break line on one side.

Therapeutic class
A12AX

Pharmacodynamic properties

Calcium citrate is the most bioavailable form of calcium. Calcium citrate formulation provides a more optimum calcium bioavailability than calcium carbonate. Calcium absorption from carbonate salts is impaired in achlorhydria. Since achlorhydria is common in older patients. Calcium carbonate is not the ideal supplement. Calcium absorption from calcium citrate is unaffected by fasting, food or achlorhydria.

Pharmacokinetic studies have concluded that the change in area under the curve (AUC) in serum calcium was 76 % higher with calcium citrate preparation as compared to calcium carbonate preparation, thereby concluding that calcium citrate preparation is much more bioavailable than calcium carbonate preparation.

Clinical studies undertaken have provided evidence that calcium citrate preparation builds more bone than calcium carbonate in young growing subjects. Long term calcium supplementations are associated with increased risk of renal calculi formation. In a long term study with calcium citrate supplementations in humans, it was noted that there was no increased propensity for crystallization of calcium in the urine.

Mechanism /mode of action

Calcium citrate is the most soluble, hence the most bioavailable form of Calcium. Since it is most soluble and the most bioavailable form of calcium, there is a greater and faster increase in bone material density in normal growing individuals and in osteoporotic patients.

Vitamin D increases the intestinal absorption of calcium.

Administration of calcium and vitamin D3 counteracts the increase of parathyroid hormone (PTH) which is caused by calcium deficiency and which causes increased bone resorption.

A clinical study of institutionalised patients suffering from vitamin D deficiency indicated that a daily intake of two tablets of calcium 500mg/vitamin D 400iu for six months normalised the value of the 25-hydroxylated metabolite of vitamin D3 and reduced secondary hyperparathyroidism and serum alkaline phosphatase.

Pharmacokinetic properties

Calcium

Absorption: The amount of calcium absorbed through the gastrointestinal tract is approximately 30% of the swallowed dose.

Distribution and metabolism: 99% of the calcium in the body is concentrated in the hard structure of bones and teeth. The remaining 1% is present in the intra- and extracellular fluids. About 50% of the total blood-calcium content is in the physiologically active ionised form with approximately 10% being complexed to citrate, phosphate or other anions, the remaining 40% being bound to proteins, principally albumin.

Elimination: Calcium is eliminated through faeces, urine and sweat. Renal excretion depends on glomerular filtration and calcium tubular reabsorption.

Vitamin D

Absorption: Vitamin D is easily absorbed in the small intestine.

Distribution and metabolism: Cholecalciferol and its metabolites circulate in the blood bound to a specific globulin. Cholecalciferol is converted in the liver by hydroxylation to the active form 25-hydroxycholecalciferol. It is then further converted in the kidneys to 1,25 dihydroxycholecalciferol; 1,25-dihydroxycholecalciferol is the metabolite responsible for increasing calcium absorption.

Vitamin D, which is not metabolised, is stored in adipose and muscle tissues.

Elimination: Vitamin D is excreted in faeces and urine.

Indications

- Pregnancy and lactation
- For treatment & prevention of postmenopausal osteoporosis.
- For prevention of osteoporosis in other susceptible populations such as the elderly, chronic bed-ridden patients: those on chronic steroid medication, anticonvulsants.
- As a therapeutic adjunct while treating established osteoporosis.

Recommended dosage

The usual dose is 1 tablet twice daily or as directed by the Physician.

Route of Administration: Oral

Adverse effects

MILICAL is generally well tolerated with an overall safety profile. In recommended dosage, **MILICAL** usually does not cause side effects such as flatulence, bloating, abdominal discomfort and constipation which are frequently encountered with calcium carbonate and other supplements.

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MILICAL relatively carries lower risk for the development of oxalate stones in the susceptible lithogenic population. Calcium citrate administration has not been demonstrated to interfere with the absorption of iron.

Excessive intake of vitamin D leads to the development of hyperphosphataemia or hypercalcaemia. Associated effects of hypercalcaemia include hypercalciuria, ectopic calcification, and renal and cardiovascular damage . Symptoms of overdose include anorexia, lassitude, nausea and vomiting, constipation or diarrhoea, polyuria, nocturia, sweating, headache, thirst, somnolence, and vertigo. Interindividual tolerance to vitamin D varies considerably; infants and children are generally more susceptible to its toxic effects. The vitamin should be withdrawn if toxicity occurs. It has been stated that vitamin D dietary supplementation may be detrimental in persons already receiving an adequate intake through diet and exposure to sunlight, since the difference between therapeutic and toxic concentrations is relatively small. Skin irritation or contact dermatitis has been reported with topical preparations..

Precautions / warnings

Amilical tablet is to be taken as per the direction in pregnant and breast feeding women.

Arteriosclerosis - potential exacerbation related to persistent hypercalcemic effects during therapeutic use.

Cardiac conditions - potential exacerbation related to persistent hypercalcemic effects during therapeutic use

Concurrent use of calcium-containing preparations, other vitamin D-containing preparations or vitamin D analogs, or thiazide diuretics will enhance the risk of hypercalcaemia.

Hyperlipidemia - potential exacerbation of LDL elevation

Hyperphosphatemia - risk of metastatic calcification; normalization of phosphate levels indicated prior to therapy

Liver disease - impaired absorption

Renal impairment - potential exacerbation related to hypercalcemic effects during therapeutic use

Renal osteodystrophy/other conditions requiring high doses of pure vitamin D (vitamin D metabolites preferred, such as calcitriol)

Sarcoidosis/other granulomatous disease - potential for enhanced sensitivity to vitamin D

Pregnancy & Lactation:

Consult physician while taking tablets during pregnancy & lactation.

Contraindications

- Hyperglycemia (secondary to primary hyperparathyroidism, Vitamin D overdose, sarcoidosis, malignant bone disease myeloma, bone metastases)
- Severe Hypercalciuria and extensive nephrolithiasis
- Severe renal failure

Drug interactions

Tetracycline antibiotic may interfere with the absorption of the calcium. There is an increased risk of hypercalcaemia if vitamin D is given with thiazide diuretics, calcium, or phosphate. Plasma-calcium concentrations should be monitored in such situations. Some antiepileptics may increase vitamin D requirements (e.g. carbamazepine, phenobarbital, phenytoin, and primidone). Rifampicin and isoniazid may reduce the effectiveness of vitamin D. Corticosteroids may counteract the effect of vitamin D.

Bran decreases the gastrointestinal absorption of calcium, and may therefore decrease the efficacy of calcium supplements. Corticosteroids also reduce calcium absorption. Calcium enhances the effects of digitalis glycosides on the heart and may precipitate digitalis intoxication; parenteral calcium therapy is best avoided in patients receiving cardiac glycosides. Citrate salts increase the absorption of aluminium from the gastrointestinal tract , therefore patients with renal failure taking aluminium compounds should avoid taking calcium citrate. Calcium salts reduce the absorption of a number of other drugs such as bisphosphonates, fluoride, some fluoroquinolones, and tetracyclines; administration should be separated by at least 3 hours.

Symptoms & treatment of Overdose:

In case of suspected overdose, consult the physician. Symptoms of overdose may include nausea, vomiting, severe drowsiness, dry mouth, loss of appetite, headache & constipation.

Storage conditions

Store below 30°C. Protect from moisture & light.

Shelf life

24 months

Pack size

Box of 3 x 10's

MAL 06061542X

Date of Revision: 02/08/2021

Manufactured By:

glenmark

PHARMACEUTICALS LTD.

Plot No. E-37, 39, D-Road,
MIDC, Satpur, Nashik - 422 007,
Maharashtra State, India.

TM Trade Mark

Product Registration Holder:

Glenmark Pharmaceuticals (Malaysia)
Sdn Bhd (660397-W)
D-31-02, Menara Suezcap 1,
No. 2, Jalan Kerinchi,
59200 Kuala Lumpur, Malaysia

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ICONGRAPHICS CODE:

PANTONE SHADE



PANTONE
BLACK
PROCESS C

**Supersedes
Artwork Code:
PE33869**

PHARMACODE: